EXCELLENCE

CLASS 11 - 2026-27

Physics | Vector 05

Vectors

Cross Product for Physics

By - Excellence Programme

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Topics to be covered

01 Why physics needs A × B

02 Magnitude A B sin θ

03 Right-hand rule (direction)

04 Dot vs cross — memory

05 $\hat{i}\,\hat{j}\,\hat{k}$ cycle

06 xy-plane shortcut

07 Torque $\tau = r \times F$

08 Board practice

Physics, not maths chapter

We only keep what Class 11 Physics actually uses.

Keep: size A B sin θ, right-hand rule, $\hat{i}\hat{j}\hat{k}$, torque language.

Skip today: triangle/parallelogram area drills, long 3×3 algebra.

Cross product gives a vector perpendicular to both factors.

Later chapters use the same idea for torque and magnetic force.

Why physics needs A × B

Dot product answered: how much along the same line?

Cross product answers: how much perpendicular turning effect?

A × B is perpendicular to both A and B

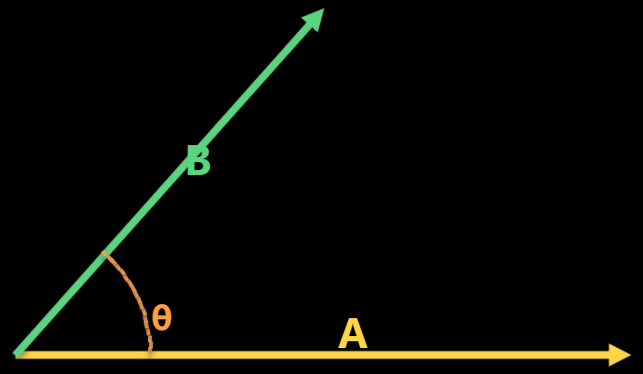
Think: A and B on a table. A × B stands straight up from the table.

If A and B are parallel, A × B = 0. No unique upright direction.

Magnitude first: A B sin θ

Only the perpendicular part of B counts. That part is B sin θ.

$$|A \times B| = A\,B \sin\theta$$



Memory

sin for cross, cos for dot.

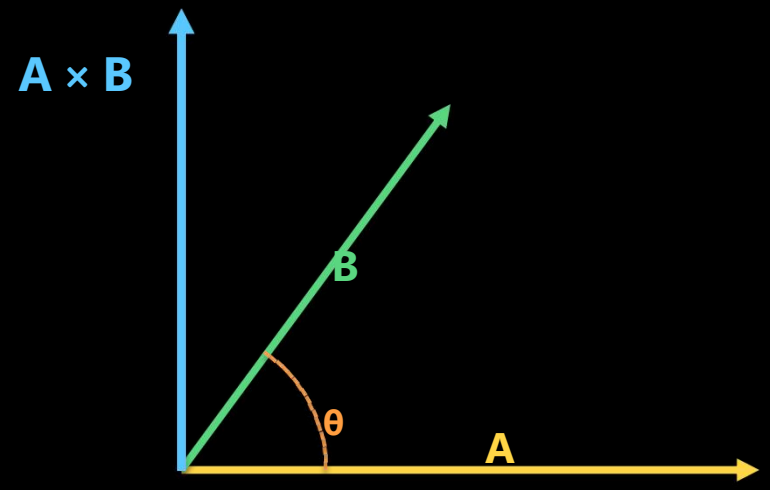
Physics habit

Max at 90°. Zero at 0° or 180°.

Rough work / extra work

Direction: right-hand rule

Fingers along A. Curl toward B through the smaller angle. Thumb = A × B.



$$A \times B = -(B \times A)$$

Order matters

Swap the arrows → result flips.

Right-hand rule is how physics fixes direction of torque.

Dot vs cross — physics memory

	Dot product	Cross product
Result	a number (scalar)	a vector
Size	$AB\cos\theta$	$AB\sin\theta$
Zero when	perpendicular	parallel
Order	$A \cdot B = B \cdot A$	$A \times B = -B \times A$
Used for	work, projection	torque, magnetic force

Cos asks how parallel. Sin asks how perpendicular.

$\hat{i}\,\hat{j}\,\hat{k}$ cycle (must know)

$$\hat{i} \times \hat{j} = \hat{k}\,,\ \hat{j} \times \hat{k} = \hat{i}\,,\ \hat{k} \times \hat{i} = \hat{j}$$

Forward in the cycle i → j → k → i: positive.

Backward: minus sign. Example: $\hat{j} \times \hat{i} = -\hat{k}$.

$$\hat{i} \times \hat{i} = 0\,,\ \hat{j} \times \hat{j} = 0\,,\ \hat{k} \times \hat{k} = 0$$

Same unit vector crossed with itself is always zero.

xy-plane shortcut (enough for now)

Most Class 11 board problems stay in the plane.

If $A = A_x \hat{i} + A_y \hat{j}$ and $B = B_x \hat{i} + B_y \hat{j}$:

$$\mathbf{A \times B = (A_x B_y - A_y B_x)\,\hat{k}}$$

Result points along $\pm \hat{k}$ only — out of the plane.

Do not start heavy 3×3 algebra unless a problem forces 3D.

Torque is a cross product

$$\tau = r \times F$$

So $|\tau| = r F \sin\theta$. Same sin pattern as $|A \times B|$.

Direction of τ: right-hand rule with fingers $r \rightarrow F$.

Max torque

Force perpendicular to r.

Zero torque

Force parallel to r ($\theta = 0°$ or $180°$).

Rough work / extra work

Where else this idea appears

1. Torque: turning effect of a force about a point.

2. Later magnetism: $F = q\,(v \times B)$ uses the same rule.

3. Angular momentum language: $L = r \times p$ (same shape).

Learn the tool once. Physics reuses it many times.

Today: master size + direction. Applications grow later.

Find |A × B|

A = 3 and B = 4. Angle between them is 30°. Find |A × B|.

Solution: Find |A × B|

Step 1: $|A \times B| = A\,B \sin\theta = 3 \times 4 \times \sin 30°$

Step 2: $\sin 30° = 1/2$, so $|A \times B| = 6$

$|A \times B| = 6$.

$\hat{\imath} \times \hat{\jmath}$ and $\hat{\jmath} \times \hat{\imath}$

Find $\hat{\imath} \times \hat{\jmath}$ and $\hat{\jmath} \times \hat{\imath}$.

Solution: $\hat{i} \times \hat{j}$ and $\hat{j} \times \hat{i}$

Step 1: $\hat{i} \times \hat{j} = \hat{k}$

Step 2: $\hat{j} \times \hat{i} = -(\hat{i} \times \hat{j}) = -\hat{k}$

$\hat{k}$ and $-\hat{k}$. Order flips the sign.

Rough work / extra work

Cross product in the xy-plane

$\mathbf{A} = 2\,\hat{\imath} + 3\,\hat{\jmath}$ and $\mathbf{B} = 4\,\hat{\imath} - \hat{\jmath}$. Find $\mathbf{A} \times \mathbf{B}$.

Solution: Cross product in the xy-plane

Step 1: $\vec{A} \times \vec{B} = (A_x B_y - A_y B_x)\,\hat{k}$

Step 2: $= (2)(-1) - (3)(4) = -2 - 12 = -14$

$\vec{A} \times \vec{B} = -14\,\hat{k}$.

When is A × B zero?

A = 2 î + 4 ĵ and B = î + 2 ĵ. Find A × B. What does that mean physically?

Solution: When is A × B zero?

Step 1: $A \times B = (2\times2 - 4\times1)\,\hat{k} = 0$

Step 2: B is parallel to A, so there is no perpendicular turning part

$A \times B = 0$. Parallel vectors → zero cross product.

Torque magnitude

A force F = 10 N acts at r = 0.4 m from the pivot. Angle between r and F is 90°. Find |τ|.

Solution: Torque magnitude

Step 1: $|\tau| = r F \sin\theta = 0.4 \times 10 \times \sin 90^\circ$

Step 2: $\sin 90^\circ = 1$, so $|\tau| = 4$ N m

$|\tau| = 4$ N m.

Rough work / extra work

When is torque zero?

Same $r = 0.4$ m and $F = 10$ N, but force is along the line of r ($\theta = 0°$). Find $|\tau|$.

Solution: When is torque zero?

Step 1: $|\tau| = r F \sin 0° = 0$

Step 2: Force through the pivot line gives no turning effect

$|\tau| = 0$. Parallel → no torque.

Maximum |A × B|

|A| = 6 and |B| = 5. What is the maximum possible |A × B|?

Solution: Maximum |A × B|

Step 1: $|A \times B| = A\,B \sin\theta \le A\,B$

Step 2: Maximum when $\theta = 90°$ (fully perpendicular)

Maximum $|A \times B| = 30$.

Rough work / extra work

Direction of torque

r is along +x (î) and F is along +y (ĵ). Direction of τ = r × F is

1. +x
2. +y
3. +z (k̂)
4. −z (−k̂)

Solution: Direction of torque

Step 1: $\hat{i} \times \hat{j} = \hat{k}$

Step 2: Right-hand rule: fingers x → y, thumb out of page (+z)

Answer: +z ($\hat{k}$).

Unit vector along A × B

A = î + ĵ and B = î − ĵ. Find a unit vector along A × B.

Solution: Unit vector along A × B

Step 1: $\vec{A} \times \vec{B} = (1\times(-1) - 1\times1)\,\hat{k} = -2\,\hat{k}$

Step 2: Unit vector = $(-2\,\hat{k})/2 = -\,\hat{k}$

Unit vector = $-\,\hat{k}$.

If $A \cdot B = 0$, what about $|A \times B|$?

If A and B are perpendicular and non-zero, $|A \times B|$ equals

1. 0
2. A B
3. A + B
4. A / B

Solution: If A · B = 0, what about |A × B|?

Step 1: $\theta = 90° \Rightarrow \sin\theta = 1 \Rightarrow |A \times B| = A\,B$

Answer: A B. (Same idea as max torque.)

Rough work / extra work

Vector 05 lock lines

$|\mathbf{A} \times \mathbf{B}| = A\,B \sin\theta$. Max at 90°. Zero if parallel.

Direction by right-hand rule. $\mathbf{A} \times \mathbf{B} = -\,\mathbf{B} \times \mathbf{A}$.

$\hat{\imath} \times \hat{\jmath} = \hat{k}$, $\hat{\jmath} \times \hat{k} = \hat{\imath}$, $\hat{k} \times \hat{\imath} = \hat{\jmath}$

In the plane: $\mathbf{A} \times \mathbf{B} = (A_x B_y - A_y B_x)\,\hat{k}$

Physics link: $\tau = r \times F$, so $|\tau| = r\,F \sin\theta$.

Practice

1. Find $|A \times B|$ if $A = 5$, $B = 8$, $\theta = 90°$.

2. Write $\hat{i} \times \hat{j}$, $\hat{j} \times \hat{k}$, $\hat{k} \times \hat{i}$, and $\hat{j} \times \hat{i}$.

3. $A = 3\,\hat{i} + \hat{j}$, $B = 2\,\hat{i} + 4\,\hat{j}$. Find $A \times B$.

4. $r = 0.5$ m, $F = 20$ N, $\theta = 30°$. Find $|\tau|$.

5. Force along the rod through the pivot: what is torque? Why?

6. r along $\hat{j}$, F along $-\hat{i}$. Direction of τ?

Dot uses cos. Cross uses sin. Torque uses the same sin.